

Number Theory Foundations

Shef Scholars Competitive Math Academy

Week 0

Introduction

This handout covers the number theory foundations that we will assume throughout the AMC 10 Get 100 Camp.

Most of the ideas in Week 0 should not be completely new. You should already have seen even and odd numbers, prime numbers, factor trees, and at least the divisibility rules for 2, 5, and 10. Our goal is to make sure that these skills are reliable enough that they do not become the reason you miss an AMC 10 problem.

That distinction matters. Number theory on the AMC 10 rarely asks you to do something you have never seen. It asks you to notice that a problem about prices, or probabilities, or a calendar, is secretly a problem about which numbers divide which. A student who knows the divisibility rule for 11 but does not recognise when to reach for it will still lose the point.

So Week 0 has two purposes:

1. **Review the number theory we will assume you know.** If something here feels unfamiliar, this is the time to identify it and learn it before the main camp begins.
2. **Learn to recognize these basic ideas when they are disguised inside AMC problems.** The number theory may be elementary even when the problem does not look elementary.

You should aim to finish Week 0 able to factor a number quickly and correctly, apply every divisibility test without looking it up, and recognize which basic tool a problem is asking you to use.

Main idea/Fact (What Week 0 is for)

Week 0 is prerequisite review, not the main number theory curriculum of the camp.

The goal is not merely to know these ideas. The goal is to be able to use them quickly, accurately, and without them getting in the way of the harder mathematics that comes later.

Divisibility Rules

What is Divisibility?

Divisibility is a fundamental concept in mathematics. Simply put, one number is divisible by another if, when you divide them, the result is a whole number (no remainder). For example, 12 is divisible by 3 because $12 \div 3 = 4$, a whole number. However, 13 is not divisible by 3 because $13 \div 3 = 4$ with a remainder of 1.

Understanding divisibility helps you quickly determine factors of numbers and is essential for solving various math problems, especially in competitions.

Divisibility Rules

Divisibility by 1

Every number is divisible by 1 because any number can be expressed as 1 times itself.

Example 1

$7 \div 1 = 7$, so 7 is divisible by 1.

Non-example: There is no number not divisible by 1.

Divisibility by 2

A number is divisible by 2 if its last digit is even, meaning the last digit is one of 0, 2, 4, 6, 8.

Example 2

18 ends in 8, so it is divisible by 2 and $18 \div 2 = 9$.

Non-example: 17 ends in 7, which is odd, so it is not divisible by 2 and $17 \div 2 = 8.5$.

Divisibility by 3

A number is divisible by 3 if the sum of its digits is divisible by 3.

Example 3

The digits of 123 sum to $1+2+3 = 6$, and $6 \div 3 = 2$, so 123 is divisible by 3 ($123 \div 3 = 41$).

Non-example: The digits of 124 sum to $1 + 2 + 4 = 7$, and $7 \div 3 = 2.\overline{33}$, so 124 is not divisible by 3 ($124 \div 3 = 41.\overline{33}$).

Divisibility by 4

A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

Example 4

The last two digits of 316 are 16, and $16 \div 4 = 4$, so 316 is divisible by 4 ($316 \div 4 = 79$).

Non-example: The last two digits of 318 are 18, and $18 \div 4 = 4.5$, so 318 is not divisible by 4 ($318 \div 4 = 79.5$).

Divisibility by 5

A number is divisible by 5 if its last digit is 0 or 5.

Example 5

40 ends in 0, so it is divisible by 5 ($40 \div 5 = 8$).

Non-example: 42 ends in 2, so it is not divisible by 5 ($42 \div 5 = 8.4$).

Divisibility by 6

A number is divisible by 6 if it is divisible by 2 *and* divisible by 3.

Example 6

2436 is divisible by both 2 (ends in 6) and 3 ($2 + 4 + 3 + 6 = 15$, $15 \div 3 = 5$), so it is divisible by 6 ($2436 \div 6 = 806$).

Non-example: 34 is divisible by 2 but $3 + 4 = 7$, and $7 \div 3 = 2.\overline{33}$, which is not divisible by 3, so 34 is not divisible by 6 ($34 \div 6 = 5.\overline{66}$).

Divisibility by 7

There is no simple, universally used divisibility rule for 7, though some special rules exist.

Example 7

$35 \div 7 = 5$, so 35 is divisible by 7.

Non-example: $37 \div 7 = 5.285\dots$, so 37 is not divisible by 7.

Divisibility by 8

A number is divisible by 8 if the number formed by its last three digits is divisible by 8.

Example 8

1,000 ends in 000, and $000 \div 8 = 0$, so it is divisible by 8 ($1000 \div 8 = 125$).

Non-example: 1,002 ends in 002, and $2 \div 8 = 0.25$, so it is not divisible by 8 ($1002 \div 8 = 125.25$).

Divisibility by 9

A number is divisible by 9 if the sum of its digits is divisible by 9.

Example 9

The digits of 729 sum to $7 + 2 + 9 = 18$, and $18 \div 9 = 2$, so it is divisible by 9 ($729 \div 9 = 81$).

Non-example: The digits of 27731 sum to $2 + 7 + 7 + 3 + 1 = 20$ and $20 \div 9 = 2.\overline{22}$, so 27731 is not divisible by 9 ($27731 \div 9 = 3081.\overline{22}$).

Divisibility by 10

A number is divisible by 10 if its last digit is 0. (Alternatively, if the number is divisible by both 2 and 5, it is divisible by 10.)

Example 10

120 ends in 0, so it is divisible by 10 ($120 \div 10 = 12$)

Non-example: 123 ends in 3, so it is not divisible by 10 ($123 \div 10 = 12.3$).

Divisibility by 11

To test divisibility by 11, write plus (+) and minus (−) signs alternately above each digit, starting with plus over the first digit from the left. Then add all digits marked with (+) and subtract the sum of all digits marked with (−). If the resulting number is divisible by 11 (including zero), the original number is divisible by 11.

Example 11

Check if 121 is divisible by 11:

$$\begin{array}{ccccc} + & - & + & & \\ 1 & 2 & 1 & & \end{array} \Rightarrow (1 + 1) - 2 = 2 - 2 = 0.$$

Since the result is 0 which is divisible by 11 ($0 \div 11 = 0$), 121 is divisible by 11.

Non-example: Is the number 918273 divisible by 11?

$$\begin{array}{cccccc} + & - & + & - & + & - \\ 9 & 1 & 8 & 2 & 7 & 3 \end{array} \Rightarrow (9 + 8 + 7) - (1 + 2 + 3) = 24 - 6 = 18.$$

Since 18 is not divisible by 11, 918273 is not divisible by 11.

Zero is divisible by every natural number.

This is because $0 \div n = 0$ for every natural number n .

Divisibility by bigger numbers

How would you check if a number is divisible by a number like 30 or 45? You break it down into the *powers of its prime factors* and check divisibility by each of those factors.

Divisibility by 30:

We factor $30 = 2^1 \cdot 3^1 \cdot 5^1$. So a number is divisible by 30 if it is divisible by 2, 3, and 5.

Example 1: Is 120 divisible by 30?

- Check divisibility by 2: 120 ends in 0 (even), so $120 \div 2 = 60$.
- Check divisibility by 3: $1 + 2 + 0 = 3$, and $3 \div 3 = 1$.
- Check divisibility by 5: 120 ends in 0, so $120 \div 5 = 24$.

Since 120 is divisible by 2, 3, and 5, it is divisible by 30. Indeed, $120 \div 30 = 4$.

Example 2: Is 45 divisible by 30?

- Check divisibility by 2: 45 ends in 5 (odd), so 45 is **not** divisible by 2.
- Check divisibility by 3: $4 + 5 = 9$, and $9 \div 3 = 3$.
- Check divisibility by 5: 45 ends in 5, and $45 \div 5 = 9$.

Since 45 is not divisible by 2, it is **not** divisible by 30.

Divisibility by 45:

We factor $45 = 3^2 \cdot 5$. So a number is divisible by 45 if it is divisible by both 9 (because $3^2 = 9$) and 5.

Example 1: Is 45135 divisible by 45?

- Sum of digits: $4 + 5 + 1 + 3 + 5 = 18$, and 18 is divisible by 9, so 45135 is divisible by 9 ($45135 \div 9 = 5015$).
- 45135 ends in 5, so it is divisible by 5 ($45135 \div 5 = 9027$).

Since 45135 is divisible by 9 and 5, it is divisible by 45:

$$45135 \div 45 = 1003$$

Example 2: Is 12345 divisible by 45?

- Divisibility by 5: 12345 ends in 5, so divisible by 5: $12345 \div 5 = 2469$.
- Divisibility by 9: Sum of digits is $1 + 2 + 3 + 4 + 5 = 15$, and $15 \div 9 = 1.\overline{66}$, so **not** divisible by 9.

Since 12345 is **not** divisible by 9, it is **not** divisible by 45.

Examples**Example 12**

Determine the unknown digits a and b in the number $\overline{3a27b}$ so that the number is divisible by 99.

We want the number

$$\overline{3a27b}$$

to be divisible by 99.

The first thing to notice is that

$$99 = 3^2 \cdot 11 = 9 \cdot 11$$

So instead of trying to divide the whole number by 99 directly, we check divisibility by 9 and divisibility by 11 separately.

Step 1. Use divisibility by 9.

A number is divisible by 9 if the sum of its digits is divisible by 9.

For the number $\overline{3a27b}$, the digits are

$$3, a, 2, 7, b$$

So the sum of the digits is

$$3 + a + 2 + 7 + b = 12 + a + b$$

Since a and b are digits, each of them is between 0 and 9. Therefore the smallest possible value of $a + b$ is 0, and the largest possible value is 18.

That means

$$12 \leq 12 + a + b \leq 30$$

Now we only care about multiples of 9 in this range. The multiples of 9 between 12 and 30 are

$$18 \quad \text{and} \quad 27$$

Therefore we must have either

$$12 + a + b = 18$$

or

$$12 + a + b = 27$$

This gives us two possibilities:

$$a + b = 6$$

or

$$a + b = 15$$

So divisibility by 9 tells us:

$$\boxed{a + b = 6 \quad \text{or} \quad a + b = 15}$$

Step 2. Use divisibility by 11.

For divisibility by 11, we take the alternating sum of the digits. For $\overline{3a27b}$, this is

$$3 - a + 2 - 7 + b$$

Now simplify:

$$3 - a + 2 - 7 + b = b - a - 2$$

For the number to be divisible by 11, this alternating sum must be divisible by 11.

So we need

$$b - a - 2$$

to be a multiple of 11.

Before doing casework, we should understand how large or small this expression can be.

The smallest possible value happens when b is as small as possible and a is as large as possible:

$$b = 0, \quad a = 9$$

This gives

$$b - a - 2 = 0 - 9 - 2 = -11$$

The largest possible value happens when b is as large as possible and a is as small as possible:

$$b = 9, \quad a = 0$$

This gives

$$b - a - 2 = 9 - 0 - 2 = 7$$

So we know

$$-11 \leq b - a - 2 \leq 7$$

The only multiples of 11 in this range are

$$-11 \quad \text{and} \quad 0$$

Therefore divisibility by 11 tells us:

$$\boxed{b - a - 2 = -11 \quad \text{or} \quad b - a - 2 = 0}$$

Step 3. Combine the two tests.

From divisibility by 9, we know

$$a + b = 6 \quad \text{or} \quad a + b = 15$$

From divisibility by 11, we know

$$b - a - 2 = -11 \quad \text{or} \quad b - a - 2 = 0$$

So we need to check four possible combinations.

Case 1. Suppose

$$a + b = 6 \quad \text{and} \quad b - a - 2 = 0$$

The second equation gives

$$b - a - 2 = 0$$

so

$$b = a + 2$$

Substitute this into $a + b = 6$:

$$a + (a + 2) = 6$$

so

$$2a + 2 = 6$$

and therefore

$$2a = 4$$

Thus

$$a = 2$$

Then

$$b = a + 2 = 2 + 2 = 4$$

So this case gives

$$(a, b) = (2, 4)$$

This is valid because both 2 and 4 are digits.

Case 2. Suppose

$$a + b = 6 \quad \text{and} \quad b - a - 2 = -11$$

The second equation gives

$$b = a - 9$$

Substitute this into $a + b = 6$:

$$a + (a - 9) = 6$$

so

$$2a - 9 = 6$$

and therefore

$$2a = 15$$

This gives

$$a = 7.5$$

But a must be a digit, so a must be an integer. Therefore this case does not work.

Case 3. Suppose

$$a + b = 15 \quad \text{and} \quad b - a - 2 = 0$$

The second equation gives

$$b = a + 2$$

Substitute this into $a + b = 15$:

$$a + (a + 2) = 15$$

so

$$2a + 2 = 15$$

and therefore

$$2a = 13$$

This gives

$$a = 6.5$$

Again, a is not an integer, so this case does not work.

Case 4. Suppose

$$a + b = 15 \quad \text{and} \quad b - a - 2 = -11$$

The second equation gives

$$b = a - 9$$

Substitute this into $a + b = 15$:

$$a + (a - 9) = 15$$

so

$$2a - 9 = 15$$

and therefore

$$2a = 24$$

This gives

$$a = 12$$

But a must be a digit from 0 to 9, so this case does not work.

Step 4. Conclude.

The only case that worked was

$$(a, b) = (2, 4)$$

Therefore

$$a = 2, \quad b = 4$$

So the number is

$$\overline{32274}$$

We can quickly check:

The digit sum is

$$3 + 2 + 2 + 7 + 4 = 18$$

so the number is divisible by 9.

The alternating sum is

$$3 - 2 + 2 - 7 + 4 = 0$$

so the number is divisible by 11.

Therefore $\overline{32274}$ is divisible by both 9 and 11, so it is divisible by 99.

Hence the answer is

$$\boxed{a = 2, \ b = 4}$$

Example 13

Determine the difference between the smallest six-digit number divisible by 72 and the smallest six-digit number divisible by 40.

We start off by looking at the numbers that are close to the smallest six-digit number. The smallest six-digit number is 100000.

Now to figure out what the smallest number divisible by any number is, a good first step is to figure out the quotient and the remainder when we divide the number by the quotient. We will do this for both numbers and then figure out the difference between the two.

Step 1. Find the smallest six-digit number divisible by 72.

We do long division of 100000 and 72:

$$\begin{array}{r}
 1388.\bar{8} \\
 72 \overline{)100000.0} \\
 \underline{72} \\
 280 \\
 \underline{216} \\
 640 \\
 \underline{576} \\
 640 \\
 \underline{576} \\
 64.0 \\
 \underline{57.6} \\
 6.4
 \end{array}$$

$$100000 = 72 \times 1388 + 64$$

Quotient: 1388 **Remainder:** 64

So we get that we have 1388 as the quotient and 64 as the remainder. This means that we need $72 - 64 = 8$ more than 100000 to get to a number divisible by 72. Another way to look at this is we can take the ceiling (next whole number) of the who quotient ($1388.\bar{8}$), which is 1389. This give us:

$$1389 \times 72 = 100008.$$

So, the smallest six-digit number divisible by 72 is 100008.

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Step 2. Find the smallest six-digit number divisible by 40.

Divide 100000 by 40:

$$100000 \div 40 = 2500.$$

This is an exact whole number, so:

$$2500 \times 40 = 100000.$$

Thus, the smallest six-digit number divisible by 40 is 100000.

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Step 3. Calculate the difference.

$$100008 - 100000 = 8.$$

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Answer. The difference is 8.

Problems

1. Determine the possible values of the unknown digit x in the number $\overline{52x4}$ so that the number is divisible by 3.

([Hint 1](#), [Hint 2](#))

2. Determine all possible values of unknown digits a and b in the number $\overline{a2024b}$ ($a \neq 0$) so that the number is divisible by 8.

([Hint 1](#), [Hint 2](#))

3. Determine all possible values of unknown digits x , y , and z so that the number $\overline{2x6y575z}$ is divisible by 55. ([Hint 1](#), [Hint 2](#))

4. Find the smallest natural number whose digits consist only of fives and which is divisible by 45.

([Hint 1](#), [Hint 2](#))

5. Which digits should be removed from the number 127912345 to obtain the largest possible number divisible by 6?

([Hint 1](#), [Hint 2](#))

6. (2016 AMC 8 #24) The digits 1, 2, 3, 4, and 5 are each used once to write a five-digit number $PQRST$. The three-digit number PQR is divisible by 4, the three-digit number QRS is divisible by 5, and the three-digit number RST is divisible by 3. What is P ?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

([Hint 1](#), [Hint 2](#))

7. (2017 AMC 8 #7) Let Z be a 6-digit positive integer, such as 247247, whose first three digits are the same as its last three digits taken in the same order. Which of the following numbers must also be a factor of Z ?

(A) 11 (B) 19 (C) 101 (D) 111 (E) 1111

([Hint 1](#), [Hint 2](#))

8. The six-digit number $\overline{237x60}$ must be divisible by 2, 3, and 5. Find the largest possible value of x .

([Hint 1](#), [Hint 2](#))

9. (2025 AMC 10B #8) Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was $\$ \underline{A} \underline{B} \underline{B} \underline{B} \underline{A}$, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?

(A) 7 (B) 8 (C) 11 (D) 14 (E) 15

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** For divisibility by 3, we do not need to divide the whole number. We only need to look at the sum of the digits.

The digit sum of $\overline{52x4}$ is

$$5 + 2 + x + 4$$

What values of x make this sum divisible by 3?

2. **MAIN IDEA:** For divisibility by 8, only the last three digits matter. So in the number $\overline{a2024b}$, the digit a does not affect divisibility by 8.

The last three digits form the number

$$\overline{24b}$$

Which digit values of b make $\overline{24b}$ divisible by 8?

3. **MAIN IDEA:** Since

$$55 = 5 \cdot 11$$

the number must be divisible by both 5 and 11.

Divisibility by 5 tells you something about the last digit z . Divisibility by 11 tells you to compare alternating digit sums. What are the possible values of z ?

4. **MAIN IDEA:** A number is divisible by 45 if it is divisible by both 5 and 9, since

$$45 = 5 \cdot 9$$

A number made only of fives is automatically divisible by 5. So the real condition comes from divisibility by 9. If there are n fives, what is the sum of the digits?

5. **MAIN IDEA:** A number is divisible by 6 if it is divisible by both 2 and 3.

To get the largest possible number, we want to remove as few digits as possible. Since the final number must be even, its last digit must be even. Which digit should be the last digit of the final number?

6. **MAIN IDEA:** Use the divisibility rules from the endings of the smaller numbers.

Since QRS is divisible by 5, its last digit S must be 5. Since PQR is divisible by 4, the last two digits QR must form a number divisible by 4. What possible pairs can QR be?

7. **MAIN IDEA:** Write the number using variables. If the first three digits are a, b, c , then the number looks like

$$abcabc$$

For divisibility by 11, compare the alternating sums of the digits. What are the two alternating sums for $abcabc$?

8. **MAIN IDEA:** The number $237x60$ already ends in 0, so it is automatically divisible by 2 and by 5.

Therefore, the only condition left is divisibility by 3. What is the digit sum of $237x60$?

9. **MAIN IDEA:** Since there are 36 shirts, the total cost in cents must be divisible by 36.

The total cost

$$\$ABB.BA$$

becomes the five-digit number of cents

$$ABBB A$$

Since $36 = 4 \cdot 9$, what divisibility rules should you apply to $ABBB A$?

Hint 2

1. The digit sum is

$$5 + 2 + x + 4 = 11 + x$$

Since x is a digit, try $x = 0, 1, 2, \dots, 9$ and keep only the values where $11 + x$ is divisible by 3. Which values work?

2. The last three digits form

$$\overline{24b} = 240 + b$$

Since 240 is already divisible by 8, we only need b itself to make the whole number divisible by 8. Which digits b are divisible by 8?

3. Divisibility by 5 gives

$$z = 0 \quad \text{or} \quad z = 5$$

For divisibility by 11, compare alternating sums:

$$(2 + 6 + 5 + 5) - (x + y + 7 + z)$$

This simplifies to

$$11 - (x + y + z)$$

So $x + y + z$ must make this expression a multiple of 11. What possible sums can $x + y + z$ have?

4. If the number has n digits, and every digit is 5, then the digit sum is

$$5n$$

For divisibility by 9, we need $5n$ to be divisible by 9. Since 5 and 9 have no common factor, what is the smallest possible value of n ?

5. To make the number even and as large as possible, keep the digit 4 as the final digit and remove the final 5.

Now the remaining number is

$$12791234$$

Its digit sum is

$$1 + 2 + 7 + 9 + 1 + 2 + 3 + 4 = 29$$

To make the digit sum divisible by 3, you need to remove a digit that leaves a multiple of 3. Which removable digit gives the largest possible final number?

6. Since $S = 5$, the pair QR must use two of the digits 1, 2, 3, 4 and must form a two-digit number divisible by 4.

The possible values are

$$QR = 12, \quad 24, \quad 32$$

Now check the condition that RST is divisible by 3. For each possible R , which remaining value of T makes $R + S + T$ divisible by 3?

7. The alternating sums of $abcabc$ are

$$a + c + b$$

and

$$b + a + c$$

These are equal, so their difference is 0. Since 0 is divisible by 11, what answer choice must always be a factor of Z ?

8. The digit sum is

$$2 + 3 + 7 + x + 6 + 0 = 18 + x$$

We need $18 + x$ to be divisible by 3. Since 18 is already divisible by 3, what digit values of x work, and which one is largest?

9. The number of cents is

$$ABBB A$$

Divisibility by 4 depends on the last two digits:

$$BA$$

Divisibility by 9 depends on the digit sum:

$$A + B + B + B + A = 2A + 3B$$

So we need BA divisible by 4 and $2A + 3B$ divisible by 9. Which digit pair (A, B) satisfies both conditions?

Note that A must be divisible by 3 because 3 which divides 9 divides $2A + 3B$ and so it must divide $2A$. Also note that if BA is divisible by 4 that A must be even.

This forces A to be a specific number and then leaves you with a few options for B .

Parity

Introduction

Divisibility by 2 is the simplest rule in Part 1, and it is also the most powerful. The reason is that “even or odd” is a single bit of information which survives addition and multiplication in a completely predictable way. That means you can often settle a question about an enormous computation without ever doing the computation.

Example 14 (2020 AMC 10A, Problem 18)

Let (a, b, c, d) be an ordered quadruple of not necessarily distinct integers, each one of them in the set $\{0, 1, 2, 3\}$. For how many such quadruples is it true that $a \cdot d - b \cdot c$ is odd? (For example, $(0, 3, 1, 1)$ is one such quadruple, because $0 \cdot 1 - 3 \cdot 1 = -3$ is odd.)

(A) 48 (B) 64 (C) 96 (D) 128 (E) 192

Solution. There are $4^4 = 256$ quadruples, so testing them is out of the question. Instead, ask when a difference is odd.

Odd minus even is odd, and even minus odd is odd, while odd minus odd and even minus even are both even. So $ad - bc$ is odd exactly when one of ad and bc is odd and the other is even.

Now, when is a product odd? Only when both of its factors are odd, since a single even factor drags in a factor of 2. The set $\{0, 1, 2, 3\}$ contains two odd values and two even ones, so among the 16 ordered pairs (a, d) exactly

$$2 \cdot 2 = 4$$

give an odd product, and the remaining 12 give an even one. The pair (b, c) behaves identically.

So we count the two cases and add:

$$\underbrace{4 \cdot 12}_{ad \text{ odd, } bc \text{ even}} + \underbrace{12 \cdot 4}_{ad \text{ even, } bc \text{ odd}} = 48 + 48 = \boxed{\text{(C) } 96}.$$

Remark. Notice that we never multiplied a single pair of numbers together. The actual values of a, b, c, d were irrelevant; all that mattered was whether each one was even or odd. That is the whole point of parity.

The main habit

Main idea/Fact (Main habit for parity)

Replace every number by a single label, E or O , and do the arithmetic on the labels.

If a problem asks whether something is even, or asks for a count of odd objects, or has answer choices of mixed parity, try the labels before you try the numbers.

Even and odd

Definition

An integer n is **even** if it can be written as

$$n = 2k$$

for some integer k , and **odd** if it can be written as

$$n = 2k + 1$$

for some integer k . The **parity** of an integer is which of the two it is.

Two integers have the **same parity** if they are both even or both odd.

Every integer is exactly one of the two. In particular $0 = 2 \cdot 0$ is even, and negative integers have parity too: -4 is even and -7 is odd.

Pitfall

Two things students routinely get wrong.

First, 0 is even. It is not “neither”.

Second, parity is only defined for integers. The number $\frac{1}{2}$ is neither even nor odd, so before you argue with parity, check that the quantity you are talking about really is a whole number.

The parity tables

Everything in this part comes from two small tables, and both of them you can rebuild in five seconds from $2k$ and $2k + 1$.

Main idea/Fact (Adding and multiplying parities)

For sums (and subtraction behaves exactly the same way):

$$E + E = E, \quad E + O = O, \quad O + O = E.$$

For products:

$$E \cdot E = E, \quad E \cdot O = O, \quad O \cdot O = O.$$

Two consequences worth saying out loud:

- A product is odd **exactly when every factor is odd**. One even factor is enough to make the whole product even.
- $a + b$ and $a - b$ always have the **same parity**, for any integers a and b .

Here is the proof of one line, so you can see they are not magic. If $a = 2m + 1$ and $b = 2n + 1$ are both odd, then

$$a + b = 2m + 1 + 2n + 1 = 2(m + n + 1),$$

which is even, and

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1,$$

which is odd.

Example 15

Without computing it, decide whether $17 \cdot 24 + 35$ is even or odd.

Solution. The product $17 \cdot 24$ has an even factor, so it is even. Then

$$E + O = O,$$

so the whole expression is odd.

Main idea/Fact (Powers keep parity)

For any integer n and any exponent $k \geq 1$, the number n^k has the same parity as n , because a power is just a product of copies of n .

So n^2 is even exactly when n is even, and odd exactly when n is odd, no matter how large the exponent gets.

Exercise 16. Is $3^{100} + 7^{99}$ even or odd?¹

Long sums

In a sum with many terms, the even terms contribute nothing to the parity. Only the odd ones matter, and even they only matter through how many there are.

Main idea/Fact (Counting the odd terms)

The parity of a sum equals the parity of **the number of odd terms** in it.

So a sum of n odd numbers is odd when n is odd, and even when n is even.

Example 17

Is $1 + 2 + 3 + \cdots + 100$ even or odd?

Solution. Among 1 through 100 there are 50 odd numbers. Since 50 is even, the sum is even. (Indeed the sum is 5050.)

Example 18

Is $1 + 3 + 5 + \cdots + 99$ even or odd?

Solution. Every term is odd, and there are 50 of them, an even count, so the sum is even. (Indeed it is 2500.)

Exercise 19. Is $1! + 2! + 3! + \cdots + 100!$ even or odd?²

¹Both powers are odd, and $O + O = E$, so the sum is even.

²Only $1! = 1$ is odd; every $n!$ with $n \geq 2$ contains a factor of 2. One odd term means the sum is odd.

Same parity, and the difference of squares

The fact that $a + b$ and $a - b$ always share a parity is worth its own section, because it is the engine behind a whole family of contest problems.

Main idea/Fact (Two numbers of the same parity)

If a and b have the same parity, then $a + b$ and $a - b$ are both even, so

$$\frac{a + b}{2} \quad \text{and} \quad \frac{a - b}{2}$$

are integers. If they have opposite parity, both are odd and neither halves evenly.

In particular, in the factoring

$$b^2 - a^2 = (b - a)(b + a),$$

the two factors on the right **always share a parity**. They are either both odd, making the product odd, or both even, making the product divisible by 4.

Pitfall

A consequence you should remember: no perfect square minus another perfect square can equal a number like 2, 6, 10, 2026, or anything else that is even but not divisible by 4.

That single observation kills entire cases in difference-of-squares problems, and we will use it again in Part 4.

Exercise 20. Can $b^2 - a^2 = 30$ for integers a and b ?³

Parity as an invariant

Sometimes a problem describes a process that repeats many times and asks what is possible at the end. Look for a quantity whose parity never changes. Anything the process cannot reach is then ruled out permanently.

Main idea/Fact (Find what never changes)

If every legal move leaves the parity of some quantity alone, then that parity is the same at the end as it was at the start.

To prove something is impossible, exhibit a parity that would have to change.

Example 21

The numbers 1, 2, 3, ..., 10 are written on a board. Repeatedly, two numbers are erased and their positive difference is written in their place. After nine moves one number remains. Can it be 0?

³No. 30 is even but not divisible by 4, and $(b - a)(b + a)$ can never be.

Solution. Track the parity of the sum of everything on the board. If we erase a and b and write $|a - b|$, the sum changes by

$$|a - b| - a - b.$$

Since $|a - b|$ and $a + b$ always have the same parity, this change is always even, so the parity of the total sum never changes.

At the start the sum is

$$1 + 2 + \cdots + 10 = 55,$$

which is odd. So the final single number is odd, and in particular it can never be 0.

Parity for elimination

Even when parity does not solve a problem outright, it often halves the work.

Main idea/Fact (Cross choices off)

If you can show the answer must be even, cross out every odd answer choice, and vice versa.

If a problem asks for a count of objects, ask whether the objects pair up. Objects that pair up come in even numbers, and a leftover forces an odd count.

Example 22 (2012 AMC 10B, Problem 9)

Two integers have a sum of 26. When two more integers are added to the first two integers the sum is 41. Finally when two more integers are added to the sum of the previous four integers the sum is 57. What is the minimum number of even integers among the 6 integers?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Solution. Handle the three pairs one at a time, and ask each time what parities are forced.

The first pair sums to 26, which is even, so the two integers are both even or both odd. We are minimising even integers, so choose both odd: zero even integers so far.

The second pair sums to $41 - 26 = 15$, which is odd. An odd sum forces one even and one odd, so this pair contributes exactly one even integer, and there is no way around it.

The third pair sums to $57 - 41 = 16$, which is even, so again both can be odd: zero more.

Therefore the minimum is

(A) 1.

Pitfall

Notice the shape of that argument. An **even** sum leaves you a choice (both even or both odd), while an **odd** sum forces exactly one of each. When a problem asks you to minimise or maximise the number of even terms, hunt for the odd sums first, because those are the ones that pin you down.

Summary

- Main idea/Fact** (Parity checklist)
- Every integer is even ($2k$) or odd ($2k + 1$), and 0 is even.
 - $E + E = E$, $E + O = O$, $O + O = E$, and subtraction behaves the same way.
 - $E \cdot E = E$, $E \cdot O = O$, $O \cdot O = O$: a product is odd exactly when every factor is odd.
 - n^k has the same parity as n .
 - The parity of a sum is the parity of the number of odd terms in it.
 - $a + b$ and $a - b$ always share a parity, so $(b - a)(b + a)$ is odd or divisible by 4, never twice an odd number.
 - In a repeating process, look for a quantity whose parity never changes, and use it to rule outcomes out.
 - An even sum of two numbers leaves a choice; an odd sum forces exactly one even and one odd.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

1. (2013 AMC 10A #4) A softball team played ten games, scoring 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 runs. They lost by one run in exactly five games. In each of the other games, they scored twice as many runs as their opponent. How many total runs did their opponents score?

(A) 35 (B) 40 (C) 45 (D) 50 (E) 55

([Hint 1](#), [Hint 2](#))

2. (2022 AMC 10B #3) How many three-digit positive integers have an odd number of even digits?

(A) 150 (B) 250 (C) 350 (D) 450 (E) 550

([Hint 1](#), [Hint 2](#))

3. (2016 AMC 10A #12) Three distinct integers are selected at random between 1 and 2016, inclusive. Which of the following is a correct statement about the probability p that the product of the three integers is odd?

(A) $p < \frac{1}{8}$ (B) $p = \frac{1}{8}$ (C) $\frac{1}{8} < p < \frac{1}{3}$ (D) $p = \frac{1}{3}$ (E) $p > \frac{1}{3}$

([Hint 1](#), [Hint 2](#))

4. (2015 AMC 10B #10) What are the sign and units digit of the product of all the odd negative integers strictly greater than -2015 ?

(A) It is a negative number ending with a 1.

(B) It is a positive number ending with a 1.

(C) It is a negative number ending with a 5.

(D) It is a positive number ending with a 5.

(E) It is a negative number ending with a 0.

([Hint 1](#), [Hint 2](#))

5. The numbers $1, 2, 3, \dots, 20$ are written on a board. Repeatedly, two of the numbers are erased and their *sum* is written in their place, until a single number remains. Can that final number be 209?

([Hint 1](#), [Hint 2](#))

6. Prove that the product of any four consecutive integers is even, and then decide whether the sum of any four consecutive integers must be even.

([Hint 1](#), [Hint 2](#))

7. (2019 AMC 10A #20) The numbers $1, 2, \dots, 9$ are randomly placed into the 9 squares of a 3×3 grid. Each square gets one number, and each of the numbers is used once. What is the probability that the sum of the numbers in each row and each column is odd?

(A) $\frac{1}{21}$ (B) $\frac{1}{14}$ (C) $\frac{5}{63}$ (D) $\frac{2}{21}$ (E) $\frac{1}{7}$

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** Sort the ten games into the two kinds using parity. In a game they won, their score was twice the opponent's, so their score was even.

Which five of the scores $1, 2, \dots, 10$ must therefore be the one-run losses?

2. **MAIN IDEA:** “An odd number of even digits” means exactly one even digit or exactly three. Count the choices position by position, remembering that the hundreds digit cannot be 0.

How many even and how many odd choices are there for the hundreds digit, and how many for each of the other two?

3. **MAIN IDEA:** A product of three numbers is odd exactly when all three are odd, so this is really a question about picking three odd numbers. The numbers are chosen *distinct*, so the pool shrinks after each pick.

How many of the integers from 1 to 2016 are odd?

4. **MAIN IDEA:** There are two separate questions here, and each is a parity question in disguise. The sign is decided by how many negative factors there are, and the units digit is decided by one very special factor in the list.

How many odd negative integers are strictly greater than -2015 ?

5. **MAIN IDEA:** Ask what the move does to the total of all the numbers on the board. Erasing a and b and writing $a + b$ removes $a + b$ and adds $a + b$ back.

What quantity does this process leave completely unchanged, not just unchanged in parity?

6. **MAIN IDEA:** For the product, use the rule that one even factor is enough. Among any four consecutive integers, the parities alternate.

For the sum, write the four integers as n , $n + 1$, $n + 2$, $n + 3$ and add them.

7. **MAIN IDEA:** Only the parities of the entries matter, so replace the grid by a pattern of five odd cells and four even cells. A row sums to an odd number exactly when it contains an odd number of odd entries, which here means one or three.

If the three row counts of odd entries are each 1 or 3 and they add to 5, what must the three counts be?

Hint 2

1. The five even scores 2, 4, 6, 8, 10 are the games they won, and against those the opponents scored half as much: 1, 2, 3, 4, 5. The five odd scores 1, 3, 5, 7, 9 are the one-run losses, and against those the opponents scored one more: 2, 4, 6, 8, 10.

What is the total of all ten opponent scores?

2. The hundreds digit has 4 even and 5 odd choices; each other digit has 5 of each. Exactly three even digits gives $4 \cdot 5 \cdot 5 = 100$ numbers, and exactly one even digit gives

$$4 \cdot 5 \cdot 5 + 5 \cdot 5 \cdot 5 + 5 \cdot 5 \cdot 5 = 350.$$

What is the total, and does it check against the 900 three-digit numbers?

3. Exactly 1008 of the 2016 integers are odd, so

$$p = \frac{1008}{2016} \cdot \frac{1007}{2015} \cdot \frac{1006}{2014}.$$

The first factor is exactly $\frac{1}{2}$, but compare each of the other two with $\frac{1}{2}$ before multiplying.

Is the product bigger or smaller than $\frac{1}{8}$?

4. The factors are $-1, -3, -5, \dots, -2013$, and there are 1007 of them, an odd count, so the product is negative. The product of odd numbers is odd, so it cannot end in 0, and one of the factors is -5 , so the product is a multiple of 5.

What is the only units digit an odd multiple of 5 can have?

5. The sum of all the numbers on the board never changes at all, since $a + b$ replaces a and b . It starts at

$$1 + 2 + \cdots + 20 = 210,$$

so the final number is 210.

Is 209 possible?

6. Among four consecutive integers, at least two are even, so the product has an even factor and is even. For the sum,

$$n + (n + 1) + (n + 2) + (n + 3) = 4n + 6 = 2(2n + 3).$$

Is that always even, and can it ever be divisible by 4?

7. The row counts must be 3, 1, 1 in some order, and so must the column counts. That means one entire row and one entire column are odd, which together use $3 + 3 - 1 = 5$ cells, exactly the number of odd values available. There are $3 \cdot 3 = 9$ ways to choose that row and column, out of $\binom{9}{5} = 126$ ways to place the odd numbers.

What does $\frac{9}{126}$ simplify to?

Primes, Composites, and Factor Trees

Introduction

Primes are the atoms of the whole numbers. Everything else in this handout is built from them, so this part is about two practical skills: deciding whether a number is prime, and breaking a number that is not prime into its primes.

Here is what those two skills look like when a contest puts them together with Part 1.

Example 23 (2021 Fall AMC 10A, Problem 5)

The six-digit number $\underline{2}\underline{0}\underline{2}\underline{1}\underline{0}\underline{A}$ is prime for only one digit A . What is A ?

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Solution. We are not going to test five six-digit numbers for primality. We are going to use the divisibility rules to destroy the impostors.

If A is even, the number is divisible by 2; if $A = 5$, it is divisible by 5. Either way the number is composite, since it is far larger than 2 or 5. That leaves

$$A \in \{1, 3, 7, 9\}.$$

Next, divisibility by 3. The digit sum is

$$2 + 0 + 2 + 1 + 0 + A = 5 + A.$$

For $A = 1$ this is 6 and for $A = 7$ this is 12, both multiples of 3. Those are out, leaving $A = 3$ and $A = 9$.

Finally, divisibility by 11 on the survivor 202103. The alternating sum is

$$2 - 0 + 2 - 1 + 0 - 3 = 0,$$

which is a multiple of 11, so 202103 is composite.

Only $A = 9$ survives, so the answer is

(E) 9.

Remark. We never verified that 202109 is prime. We did not have to: the problem promised exactly one digit works, and we eliminated the other four. Reading what the problem guarantees is part of the technique.

The main habit

Main idea/Fact (Main habit for primes)

Never test a number for primality by brute force before you have run the cheap divisibility rules on it.

Check 2, 3, 5, and 11 first, since each takes a glance. Only then start trial dividing, and only by primes, and only up to $\sqrt{n}$.

Prime and composite

Definition

A whole number $p > 1$ is **prime** if its only positive divisors are 1 and p itself.

A whole number $n > 1$ that is not prime is **composite**, meaning it can be written as $n = ab$ with $1 < a < n$ and $1 < b < n$.

The number 1 is neither prime nor composite.

Main idea/Fact (The primes below 100)

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47,
53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

There are 25 of them. Know this list on sight; you will use it in every part that follows.

Pitfall

Excluding 1 is not an arbitrary convention. If 1 counted as prime, then $12 = 2^2 \cdot 3$ and $12 = 1 \cdot 2^2 \cdot 3$ and $12 = 1^2 \cdot 2^2 \cdot 3$ would all be different prime factorizations, and the uniqueness we are about to rely on would collapse.

Pitfall

“Prime” does not mean “odd”. The number 2 is prime, and it is the only even prime, because every other even number has 2 as a proper divisor.

This is why so many problems begin “let p be a prime” and then quietly need you to handle $p = 2$ separately.

Exercise 24. Which of 51, 57, 61, 91 are prime?⁴

Testing whether a number is prime

Trial division works, but doing it blindly is slow. Two observations cut the work down enormously.

⁴Only 61. Note $51 = 3 \cdot 17$, $57 = 3 \cdot 19$, and $91 = 7 \cdot 13$; the last is the single most commonly misremembered “prime”.

Main idea/Fact (Stop at the square root)

To decide whether n is prime, you only need to trial divide by the **primes** p with

$$p \leq \sqrt{n}.$$

Why: divisors come in pairs d and $\frac{n}{d}$, whose product is n . If both members of a pair were larger than $\sqrt{n}$, their product would exceed n . So every composite n has a divisor at most $\sqrt{n}$, and hence a *prime* divisor at most $\sqrt{n}$.

Example 25

Is 221 prime?

Solution. Since $14^2 = 196$ and $15^2 = 225$, we have $\sqrt{221} < 15$, so we test the primes 2, 3, 5, 7, 11, 13.

It is odd, its digit sum is 5, it does not end in 0 or 5, and $221 = 7 \cdot 31 + 4$. Its alternating sum is $2 - 2 + 1 = 1$, so 11 fails. Finally

$$221 = 13 \cdot 17,$$

so 221 is composite.

Example 26

Is 397 prime?

Solution. Since $19^2 = 361$ and $20^2 = 400$, we test the primes up to 19, namely 2, 3, 5, 7, 11, 13, 17, 19.

It is odd. Its digit sum is 19, so 3 fails. It does not end in 0 or 5. Then $397 = 7 \cdot 56 + 5$, the alternating sum $3 - 9 + 7 = 1$ rules out 11, $397 = 13 \cdot 30 + 7$, $397 = 17 \cdot 23 + 6$, and $397 = 19 \cdot 20 + 17$.

Nothing divides it, so 397 is prime.

Main idea/Fact (The sieve of Eratosthenes)

To find every prime up to N at once, write out $2, 3, \dots, N$, circle 2 and cross out every larger multiple of 2, then circle the next uncrossed number and cross out its larger multiples, and repeat.

You may stop crossing out once the number you are circling exceeds $\sqrt{N}$, for exactly the reason above. Everything still uncrossed is prime.

Primes are $6k \pm 1$

Here is a shortcut that makes trial division noticeably faster, and that shows up directly in AMC problems.

Main idea/Fact (Every prime past 3 has the form $6k \pm 1$)

Divide any integer by 6. The remainder is one of 0, 1, 2, 3, 4, 5, so every integer has exactly one of the forms

$$6k, \quad 6k + 1, \quad 6k + 2, \quad 6k + 3, \quad 6k + 4, \quad 6k + 5.$$

Now look at what those forms are divisible by. The forms $6k$, $6k + 2$, $6k + 4$ are all even, and $6k + 3 = 3(2k + 1)$ is divisible by 3. A prime larger than 3 is divisible by neither 2 nor 3, so the only forms left are

$$6k + 1 \quad \text{and} \quad 6k + 5 = 6(k + 1) - 1.$$

That is, **every prime greater than 3 can be written as $6k \pm 1$.**

Check it: $5 = 6 - 1$, $7 = 6 + 1$, $11 = 12 - 1$, $13 = 12 + 1$, $17 = 18 - 1$, $19 = 18 + 1$, $23 = 24 - 1$, $29 = 30 - 1$, $31 = 30 + 1$.

Pitfall

The statement does **not** reverse. Plenty of numbers of the form $6k \pm 1$ are composite: $25 = 6 \cdot 4 + 1$, $35 = 6 \cdot 6 - 1$, $49 = 6 \cdot 8 + 1$, $55 = 6 \cdot 9 + 1$.

So $6k \pm 1$ narrows your search. It never certifies primality.

Main idea/Fact (The useful consequence)

Write a prime $p > 3$ as $p = 6k \pm 1$. Squaring,

$$p^2 = 36k^2 \pm 12k + 1, \quad \text{so} \quad p^2 - 1 = 12k(3k \pm 1).$$

So $p^2 - 1$ is divisible by 12. In fact $k(3k \pm 1)$ is always even (if k is even we are done, and if k is odd then $3k \pm 1$ is even), so

$$p^2 - 1 \text{ is divisible by } 24 \quad \text{for every prime } p > 3.$$

The headline you will actually use: **p^2 leaves a remainder of 1 when divided by 3,** for every prime p other than 3.

Example 27 (2018 AMC 10B, Problem 11)

Which of the following expressions is never a prime number when p is a prime number?

- (A) $p^2 + 16$ (B) $p^2 + 24$ (C) $p^2 + 26$ (D) $p^2 + 46$ (E) $p^2 + 96$

Solution. For every prime p except 3, we just showed p^2 leaves remainder 1 on division by 3. So $p^2 + c$ leaves remainder $1 + c$, and we want the constant c whose remainder makes the total divisible by 3, that is, c leaving remainder 2.

Check the constants: 16 leaves 1, 24 leaves 0, 26 leaves 2, 46 leaves 1, 96 leaves 0. Only choice (C) qualifies.

Now clean up the exceptional cases the argument skipped. For $p = 3$,

$$9 + 26 = 35 = 5 \cdot 7,$$

composite, and for $p = 2$,

$$4 + 26 = 30,$$

composite. So $p^2 + 26$ is always divisible by 3 or otherwise composite, and never prime:

$$\boxed{(C) \ p^2 + 26}.$$

Remark. There are infinitely many primes, a fact proved by Euclid: if $p_1, \dots, p_n$ were all of them, then $p_1 p_2 \cdots p_n + 1$ leaves remainder 1 on division by each, so its prime factors are none of them, a contradiction.

You will never need this on the AMC. It is here so that “are we going to run out of primes?” has an answer.

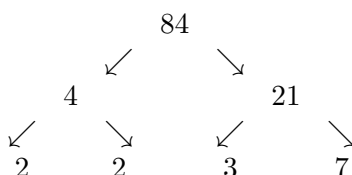
Factor trees

To break a composite number into primes, split it into any two factors, then split those, and keep going until every leaf is prime.

Example 28

Build a factor tree for 84.

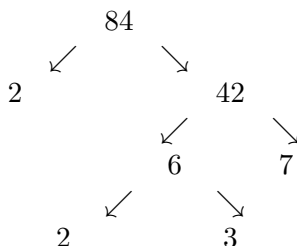
Solution. Split $84 = 4 \cdot 21$, then split each piece:



The leaves are 2, 2, 3, 7, so

$$84 = 2^2 \cdot 3 \cdot 7.$$

Now start differently, with $84 = 2 \cdot 42$:



The leaves are 2, 2, 3, 7 again. Different tree, same leaves.

Main idea/Fact (The Fundamental Theorem of Arithmetic)

Every whole number greater than 1 can be written as a product of primes, and **in exactly one way** apart from the order of the factors.

This is why the two trees above had to agree, and it is what licenses every technique in Part 4.

The Prime Factorization

Everything starts from the one form that the Fundamental Theorem of Arithmetic just promised us.

Definition 29. The **prime factorization** of a whole number $n > 1$ writes it as a product of prime powers,

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k},$$

with distinct primes $p_1 < p_2 < \cdots < p_k$ and positive exponents a_i . By the Fundamental Theorem of Arithmetic, this can be done in exactly one way.

There is a second method besides the tree, and it is the one to use when you want to be systematic: divide by the smallest prime that works, and repeat on the quotient.

Example 30

Find the prime factorization of 720.

Solution. Divide by 2 four times, $720 \rightarrow 360 \rightarrow 180 \rightarrow 90 \rightarrow 45$; then 2 stops, so divide by 3 twice, $45 \rightarrow 15 \rightarrow 5$; and 5 is prime. Collecting,

$$720 = 2^4 \cdot 3^2 \cdot 5.$$

Interpret: the only data we keep are the exponents 4, 2, 1. In general: **every tool in the next part reads this exponent list and nothing else.**

Summary

- Main idea/Fact (Primes checklist)**
- A prime has exactly two positive divisors; 1 is neither prime nor composite; 2 is the only even prime.
 - Memorize the 25 primes below 100, and remember that 51, 57, and 91 are not among them.
 - To test n , run the cheap divisibility rules first, then trial divide by primes up to $\sqrt{n}$, because divisors pair as d and $\frac{n}{d}$.
 - Every prime greater than 3 has the form $6k \pm 1$; the converse is false.
 - For every prime $p \neq 3$, the square p^2 leaves remainder 1 on division by 3 (and $p^2 - 1$ is divisible by 24 when $p > 3$).
 - A factor tree splits a number into any two factors repeatedly; by the Fundamental Theorem of Arithmetic the leaves are the same no matter how you split.
 - Write the result as $n = p_1^{a_1} \cdots p_k^{a_k}$ and keep the exponents.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

1. Build a factor tree for 504 two different ways, and write its prime factorization in the form $p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$.

([Hint 1](#), [Hint 2](#))

2. Decide whether 391 is prime, listing every prime you had to test and explaining why you could stop where you did.

([Hint 1](#), [Hint 2](#))

3. (2024 AMC 10A #3) What is the sum of the digits of the smallest prime that can be written as a sum of 5 distinct primes?

(A) 5 (B) 7 (C) 9 (D) 10 (E) 11

([Hint 1](#), [Hint 2](#))

4. (2020 AMC 10B #4) The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?

(A) 2 (B) 3 (C) 5 (D) 7 (E) 11

([Hint 1](#), [Hint 2](#))

5. (2022 AMC 10B #6) How many of the first ten numbers of the sequence 121, 11211, 1112111, ... are prime numbers?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

([Hint 1](#), [Hint 2](#))

6. (2021 Fall AMC 10B #8) The largest prime factor of 16384 is 2, because $16384 = 2^{14}$. What is the sum of the digits of the largest prime factor of 16383?

(A) 3 (B) 7 (C) 10 (D) 16 (E) 22

([Hint 1](#), [Hint 2](#))

7. (2013 AMC 10B #9) Three positive integers are each greater than 1, have a product of 27000, and are pairwise relatively prime. What is their sum?

(A) 100 (B) 137 (C) 156 (D) 160 (E) 165

([Hint 1](#), [Hint 2](#))

8. Suppose p and $p + 2$ are both primes greater than 3 (such a pair is called a pair of *twin primes*). Show that $p + 1$ must be divisible by 6.

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** Any first split works, so pick one you can see: $504 = 8 \cdot 63$ is easy, and so is $504 = 2 \cdot 252$. Keep splitting until every leaf is prime.

What are the leaves of the first tree, and do they match the leaves of the second?

2. **MAIN IDEA:** Run the cheap rules first: it is odd, its digit sum is 13, it does not end in 0 or 5. Then trial divide by primes, and stop once the primes exceed $\sqrt{391}$.

Since $19^2 = 361$ and $20^2 = 400$, which primes do you actually have to test?

3. **MAIN IDEA:** This is a parity question wearing a primes costume. Five primes are being added, and the result has to be prime, so in particular it has to be odd (it is far bigger than 2).

If one of the five primes were 2, what would the parity of the sum be?

4. **MAIN IDEA:** The two acute angles of a right triangle sum to 90, so $a + b = 90$ with both prime and $a > b$. Since 90 is even, parity tells you whether 2 can be one of them.

Now test the small primes for b in increasing order: for each one, is $90 - b$ also prime?

5. **MAIN IDEA:** Do not test these one at a time; find the pattern. The n th term has n ones, then a 2, then n ones, so write it in terms of $x = 10^n$.

The term equals $\frac{10x^2 + 9x - 1}{9}$. Can you factor $10x^2 + 9x - 1$?

6. **MAIN IDEA:** Do not trial divide a five-digit number. Notice that $16383 = 16384 - 1 = 2^{14} - 1$, which is a difference of squares.

What is $2^{14} - 1$ written as a product of two numbers?

7. **MAIN IDEA:** “Pairwise relatively prime” means no two of the three share a prime factor, so each prime in the factorization of 27000 must go *entirely* to one of the three numbers.

What is the prime factorization of 27000, and how many distinct primes does it contain?

8. **MAIN IDEA:** Use the $6k \pm 1$ classification on p . There are only two possible forms, so check them one at a time and see which is consistent with $p + 2$ also being prime.

If $p = 6k + 1$, what form does $p + 2$ have, and what is it divisible by?

Hint 2

1. From $504 = 8 \cdot 63$ the leaves are 2, 2, 2 from the 8 and 3, 3, 7 from the 63. Collect equal primes into powers.

What is 504 in the form $p_1^{a_1} p_2^{a_2} p_3^{a_3}$?

2. You only need 2, 3, 5, 7, 11, 13, 17, 19, because a composite must have a prime factor at most $\sqrt{391} < 20$. Checking down the list, $391 = 17 \cdot 23$.

Is 391 prime or composite, and what is its prime factorization?

3. Including 2 makes the sum even (one even plus four odds), and an even number bigger than 2 is never prime. So all five primes are odd, and the smallest candidates are

$$3 + 5 + 7 + 11 + 13 = 39,$$

which is $3 \cdot 13$. Bump the largest term up to the next prime.

Is $3 + 5 + 7 + 11 + 17$ prime, and what is its digit sum?

4. If one angle were 2, the other would be 88, which is even and larger than 2, so not prime. Hence both are odd primes. Now test upward: $b = 3$ gives $87 = 3 \cdot 29$, and $b = 5$ gives $85 = 5 \cdot 17$.

What does $b = 7$ give, and is it prime?

5. Factoring, $10x^2 + 9x - 1 = (10x - 1)(x + 1)$, so with $x = 10^n$ the term is

$$\frac{(10^{n+1} - 1)(10^n + 1)}{9} = \underbrace{11 \cdots 1}_{n+1} \cdot (10^n + 1).$$

For $n = 1$ this reads $11 \cdot 11 = 121$, and for $n = 2$ it reads $111 \cdot 101 = 11211$.

Since both factors exceed 1 for every $n \geq 1$, how many of the ten terms are prime?

6. By difference of squares,

$$2^{14} - 1 = (2^7 - 1)(2^7 + 1) = 127 \cdot 129 = 127 \cdot 3 \cdot 43.$$

Check that 127 is prime by testing primes up to 11.

What is the digit sum of the largest prime factor?

7. Since $27000 = 2^3 \cdot 3^3 \cdot 5^3$ and there are exactly three distinct primes, each of the three numbers gets one prime power in full: 8, 27, and 125.

What is $8 + 27 + 125$?

8. If $p = 6k + 1$ then $p + 2 = 6k + 3 = 3(2k + 1)$, which is divisible by 3 and bigger than 3, so it is not prime. Therefore $p = 6k - 1$, and then

$$p + 1 = 6k.$$

Why does that finish the problem, and what does it say about 5 and 7, or about 11 and 13?

Reading the Prime Factorization

Introduction

In the previous part we learned that every whole number is built from prime atoms. Here we learn to read them. Once a number is broken into primes, almost every question about it stops being about the number and becomes a question about the exponents in its factorization. Here is the skill at full height.

Example 31 (2021 Fall AMC 10B #6)

The least positive integer with exactly 2021 distinct positive divisors can be written in the form $m \cdot 6^k$, where m and k are integers and 6 is not a divisor of m . What is $m + k$?

(A) 47 (B) 58 (C) 59 (D) 88 (E) 90

Solution. We will prove a formula in this part: if $n = p_1^{a_1} \cdots p_k^{a_k}$, then the number of divisors of n is $(a_1 + 1) \cdots (a_k + 1)$. So we need exponents whose “plus-one” product equals 2021. Factor the target with the tools from Part 3: $2021 = 43 \cdot 47$, both prime. So our number is either one prime to the 2020, or two primes to the 42 and 46. The two-prime version is vastly smaller. To minimize it, put the larger exponent on the smaller prime: $2^{46} \cdot 3^{42}$. Now match the requested shape,

$$2^{46} \cdot 3^{42} = 2^4 \cdot (2^{42} \cdot 3^{42}) = 16 \cdot 6^{42},$$

so $m = 16$, $k = 42$, and $m + k = \boxed{\text{(B) } 58}$.

Watch what we did: we never multiplied anything out. The whole problem lived in the exponents. We factored 2021, split it into exponents, and placed them greedily.

Main idea/Fact (Main Habit)

READ THE EXPONENTS. Write every number as $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. Then divisibility, the number of divisors, and the test for a perfect square are all just statements about the exponents $a_1, a_2, \dots, a_k$. Factor first, then work entirely in the exponents.

Which Numbers Divide n ?

In Part 3 we saw a light version of this: a divisor’s atoms must fit inside the original’s atoms. Here is the full statement.

Main idea/Fact (Divisibility by Atoms)

A DIVISOR’S EXPONENTS NEVER EXCEED THE ORIGINAL’S. A number $d = p_1^{b_1} \cdots p_k^{b_k}$ divides $n = p_1^{a_1} \cdots p_k^{a_k}$ exactly when $b_i \leq a_i$ for every prime. A divisor may drop a prime entirely (that is the case $b_i = 0$), but it may never introduce a new prime: a divisor of n is built only from n ’s own primes.

Example 32

List every positive divisor of 36.

Solution. Since $36 = 2^2 \cdot 3^2$, a divisor is $2^b 3^c$ with $b \in \{0, 1, 2\}$ and $c \in \{0, 1, 2\}$. Laying out

every combination in a grid:

	3^0	3^1	3^2
2^0	1	3	9
2^1	2	6	18
2^2	4	12	36

Nine divisors in all. Interpret: each divisor sits at one cell, choosing a power of 2 for its row and a power of 3 for its column, independently. In general: **the divisors of a number are exactly the independent combinations of prime powers, each capped by the original exponent.**

Counting the Divisors

The grid above is begging to be counted. It has three rows and three columns, so $3 \times 3 = 9$ cells. That multiplication is the whole idea.

Main idea/Fact (The Number of Divisors)

ADD ONE TO EACH EXPONENT AND MULTIPLY. If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, then the number of positive divisors of n is

$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

Here is why: a divisor chooses an exponent for p_i anywhere from 0 up to a_i , which is $a_i + 1$ choices (count the fence posts: from 0 to a_i inclusive is $a_i + 1$ values, not a_i). The choices for different primes are independent, so by the multiplication principle (independent choices multiply: m ways for the first and n for the second give mn in all), we multiply.

Example 33

How many positive divisors does 360 have?

Solution. Factoring as in Part 3, $360 = 2^3 \cdot 3^2 \cdot 5^1$. The exponents are 3, 2, 1, so

$$d(360) = (3 + 1)(2 + 1)(1 + 1) = 4 \cdot 3 \cdot 2 = 24.$$

Interpret: twenty-four divisors, counted without writing a single one of them down. As a check, the grid for $36 = 2^2 \cdot 3^2$ had $(2 + 1)(2 + 1) = 9$ cells, exactly the nine divisors we listed. In general: **the divisor count is a product of (exponent-plus-one) terms, and that is the whole formula.**

The formula also runs backwards, which is where it gets powerful.

Example 34

Which positive integers have exactly three divisors?

Solution. We need $(a_1 + 1)(a_2 + 1) \cdots = 3$. Since 3 is prime, the product can only be 3 itself, coming from a single factor $a_1 + 1 = 3$. So $a_1 = 2$ and there are no other primes: the number is p^2 , a prime squared, and nothing else. Its three divisors are 1, p , and p^2 . Interpret: “exactly three divisors” is a disguised way of saying “the square of a prime.” In

general: **reading the divisor formula backwards turns a count into a shape**, the same move that cracked the opening problem and several below.

Perfect Squares and the Odd-Divisor Surprise

A perfect square is a number times itself, so squaring doubles every exponent. That single observation runs this entire section.

Main idea/Fact (Perfect Squares Have Even Exponents)

A NUMBER IS A PERFECT SQUARE EXACTLY WHEN EVERY EXPONENT IS EVEN. If $n = p_1^{a_1} \cdots p_k^{a_k}$, then n is a perfect square if and only if every a_i is even, because

$$(p_1^{c_1} \cdots p_k^{c_k})^2 = p_1^{2c_1} \cdots p_k^{2c_k}$$

has all even exponents, and conversely all even exponents let you halve them to recover a square root.

Example 35 (2016 AMC 8 #7)

Which of the following numbers is not a perfect square?

(A) 1^{2016} (B) 2^{2017} (C) 3^{2018} (D) 4^{2019} (E) 5^{2020}

Solution. A prime power is a perfect square exactly when its exponent is even. Choices A, C, and E already display even exponents. Choice D hides its base: $4^{2019} = (2^2)^{2019} = 2^{4038}$, an even exponent, so it is a square. Only 2^{2017} carries an odd exponent, so

(B) 2^{2017}

is not a perfect square. Interpret: the trap was D, where the base 4 is itself a square, so every power of it stays square. In general: **rewrite each base in primes before you judge the exponents.**

The even-exponent rule also tells us how to count the square divisors of a number.

Example 36

How many divisors of 720 are perfect squares?

Solution. We have $720 = 2^4 \cdot 3^2 \cdot 5^1$, and a square divisor needs even exponents. For the prime 2, the even values in $\{0, 1, 2, 3, 4\}$ are 0, 2, 4: three choices. For 3, the even values in $\{0, 1, 2\}$ are 0, 2: two choices. For 5, the only even value in $\{0, 1\}$ is 0: one choice. By the multiplication principle,

$$3 \cdot 2 \cdot 1 = 6$$

square divisors. Interpret: it is the same counting as before, restricted to even exponents. In general: **to count square divisors, count the even exponent choices for each prime and multiply.**

Finally, a surprise that looks like magic until you see the exponents.

Main idea/Fact (Odd Divisor Count Means Perfect Square)

$d(n)$ **IS ODD IF AND ONLY IF n IS A PERFECT SQUARE.** Two ways to see it. By the formula, $d(n) = (a_1 + 1) \cdots (a_k + 1)$ is odd exactly when every factor $(a_i + 1)$ is odd, that is, every a_i is even, that is, n is a square. By pairing (the divisor-pair idea behind the square root test in Part 3), every divisor d partners with n/d , and the two are distinct unless $d = n/d$, meaning $d = \sqrt{n}$. A non-square sorts all its divisors into distinct pairs, an even count; a square has one unpaired middle divisor $\sqrt{n}$, an odd count.

This is the secret behind the classic locker problem: after every student walks the hall flipping lockers, the ones left open are exactly the perfect-square numbers $1, 4, 9, 16, \dots$, because a locker is flipped once for each of its divisors and ends open only when it has an odd number of them.

Prime Thinking in Disguise

Many AMC problems bury a factorization question inside a story about probability, money, or counting. The skill is to spot the prime structure underneath.

Example 37 (2023 AMC 10A #14)

A number is chosen at random from among the first 100 positive integers, and a positive integer divisor of that number is then chosen at random. What is the probability that the chosen divisor is divisible by 11?

- (A) $\frac{4}{100}$ (B) $\frac{9}{200}$ (C) $\frac{1}{20}$ (D) $\frac{11}{200}$ (E) $\frac{3}{50}$

Solution. A divisor can be a multiple of 11 only if the original number is. Among 1 through 100, the multiples of 11 are $11, 22, 33, \dots, 99$: nine of them, and each carries 11 to the first power only. Now the prime-thinking step. For such a number $n = 11 \cdot t$ with $11 \nmid t$, every divisor either includes the single factor of 11 or it does not, an independent yes-or-no choice that splits the divisors into two equal halves. So given a multiple of 11, exactly half of its divisors are themselves multiples of 11: a probability of $\frac{1}{2}$. Combining with the chance of picking a multiple of 11 in the first place,

$$P = \underbrace{\frac{9}{100}}_{\text{pick a multiple of 11}} \cdot \underbrace{\frac{1}{2}}_{\text{the divisor keeps the 11}} = \frac{9}{200},$$

so the answer is **(B) $\frac{9}{200}$** . Interpret: because 11 appears to the first power, the exponent of 11 in a divisor is 0 or 1, a perfect coin flip. In general: **when a prime appears to the first power, exactly half of all divisors carry it and half do not.**

The same instinct cracks problems with no probability in sight. A total price split evenly among people must be divisible by the price per person, so factor it. A number that “leaves a remainder” or is “a multiple of” something is handing you a divisibility rule from Part 1. The problems below run through these guises.

Summary

Main idea/Fact (Number Theory Checklist)

The things to try:

- **Factor first.** Write $n = p_1^{a_1} \cdots p_k^{a_k}$; everything below reads only the exponents.
- **Divisibility.** $d \mid n$ exactly when each prime's exponent in d is at most its exponent in n , and no new primes appear.
- **Counting divisors.** $d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$; the plus-one is fence-post counting from 0 to a_i .
- **Backwards.** A required divisor count factors into the (exponent-plus-one) pieces: “three divisors” means p^2 , “four” means p^3 or pq .
- **Perfect squares.** A number is a square exactly when all exponents are even; count square divisors by counting even exponent choices and multiplying.
- **Odd count.** $d(n)$ is odd exactly when n is a perfect square, since divisors pair as d and n/d except for the lone $\sqrt{n}$.
- **Prime thinking.** A prime to the first power sits in half the divisors; learn to see disguised divisibility in word problems.

Problems

1. Find the prime factorization of 1080, and use it to determine how many positive divisors 1080 has.

([Hint 1](#), [Hint 2](#))

2. (2011 AMC 8 #17) Let w , x , y , and z be whole numbers. If $2^w \cdot 3^x \cdot 5^y \cdot 7^z = 588$, then what does $2w + 3x + 5y + 7z$ equal?

([Hint 1](#), [Hint 2](#))

3. How many integers from 1 to 200 have an odd number of positive divisors?

([Hint 1](#), [Hint 2](#))

4. (2014 AMC 8 #8) Eleven members of the Middle School Math Club each paid the same integer amount for a guest speaker. In all, they paid the guest speaker $\overline{1A2}$ dollars (a three-digit number). What is the missing digit A ?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

([Hint 1](#), [Hint 2](#))

5. What is the smallest positive integer with exactly 18 positive divisors?

([Hint 1](#), [Hint 2](#))

6. (2012 AMC 8 #18) What is the smallest positive integer that is neither prime nor a perfect square and that has no prime factor less than 50?

(A) 3127 (B) 3133 (C) 3137 (D) 3139 (E) 3149

([Hint 1](#), [Hint 2](#))

7. (2020 AMC 10A #15) A positive integer divisor of $12!$ is chosen at random. The probability that the divisor chosen is a perfect square can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

(A) 3 (B) 5 (C) 12 (D) 18 (E) 23

([Hint 1](#), [Hint 2](#))

8. (2019 AMC 10B #19) Let S be the set of all positive integer divisors of 100,000. How many numbers are the product of two distinct elements of S ?

(A) 98 (B) 100 (C) 117 (D) 119 (E) 121

([Hint 1](#), [Hint 2](#))

9. (2024 AMC 10A #5) What is the least value of n such that $n!$ is a multiple of 2024?

(A) 11 (B) 21 (C) 22 (D) 23 (E) 253

([Hint 1](#), [Hint 2](#))

10. (2024 AMC 10A #7) The product of three integers is 60. What is the least possible positive sum of the three integers?

([Hint 1](#), [Hint 2](#))

11. (2014 AMC 10B #12) The largest divisor of 2,014,000,000 is itself. What is its fifth-largest divisor?
 (A) 125,875,000 (B) 201,400,000 (C) 251,750,000 (D) 402,800,000 (E) 503,500,000
([Hint 1](#), [Hint 2](#))
12. (2017 AMC 10B #20) The number $21! = 51,090,942,171,709,440,000$ has over 60,000 positive integer divisors. One of them is chosen at random. What is the probability that it is odd?
 (A) $\frac{1}{21}$ (B) $\frac{1}{19}$ (C) $\frac{1}{18}$ (D) $\frac{1}{2}$ (E) $\frac{11}{21}$
([Hint 1](#), [Hint 2](#))
13. (2024 AMC 10A #15) Let M be the greatest integer such that both $M + 1213$ and $M + 3773$ are perfect squares. What is the units digit of M ?
 (A) 1 (B) 2 (C) 3 (D) 6 (E) 8
([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. Build the factorization by dividing by the smallest prime that works: $1080 \rightarrow 540 \rightarrow 270 \rightarrow 135$, and then 2 stops. Once you have $1080 = p_1^{a_1} \cdots p_k^{a_k}$, the divisor count adds one to each exponent and multiplies.

What are the exponents in the factorization of 1080?

2. Factor 588 into primes and match each prime's power to its variable. The number is even, so pull out the 2's first, then handle what remains.

What is 588 written as $2^w \cdot 3^x \cdot 5^y \cdot 7^z$?

3. A number has an odd number of divisors exactly when it is a perfect square, because divisors pair off as d and n/d except at $\sqrt{n}$. So you are really counting the perfect squares up to 200.

What is the largest perfect square that is at most 200?

4. Eleven people each paid a whole number of dollars, so the total $\overline{1A2}$ is a multiple of 11. Use the divisibility rule for 11 from Part 1: the alternating sum of the digits is a multiple of 11.

What equation does the alternating sum give for A ?

5. You want exponents whose plus-one product is 18. Factor 18 every way: 18 , $9 \cdot 2$, $6 \cdot 3$, and $3 \cdot 3 \cdot 2$. Each splitting gives an exponent pattern, and to make the number small you put the largest exponents on the smallest primes.

Which splitting do you expect to give the smallest number?

6. “No prime factor less than 50” forces every prime factor to be at least the smallest prime above 50. “Neither prime nor a perfect square” rules out a single prime and a prime squared.

What is the smallest prime that is at least 50?

7. First factor $12!$ into primes by counting how many times each prime appears in $1 \cdot 2 \cdots 12$ (count the multiples of the prime, then the multiples of its square, and so on). This gives $12! = 2^{10} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 11$, with $(10+1)(5+1)(2+1)(1+1)(1+1)$ divisors in total. A square divisor needs every exponent even.

How many even choices are there for the exponent of each prime?

8. Factor $100,000 = 10^5 = 2^5 \cdot 5^5$. A product of two of its divisors is $2^a \cdot 5^b$ with $0 \leq a, b \leq 10$, so it is a divisor of $100,000^2 = 2^{10} \cdot 5^{10}$.

How many numbers $2^a 5^b$ with $0 \leq a, b \leq 10$ are there in total, and which of them can only come from two equal factors?

9. Factor the target into primes: $2024 = 2^3 \cdot 11 \cdot 23$. For $n!$ to be a multiple of 2024, every one of these prime powers must already sit inside the product $1 \cdot 2 \cdots n$. The block 2^3 and the prime 11 appear early; the largest prime is the whole question.

Which prime factor of 2024 is the bottleneck?

10. The three integers are allowed to be negative. Two negatives and one positive keep the product positive while pulling the sum down. Factor 60 into three positive factors a , b , c ; negating two of them makes the sum $c - a - b$, so you want the largest factor just barely above the other two.

Which factor triple of 60 makes $c - a - b$ smallest while staying positive?

11. Divisors come in pairs d and N/d (the pairing from the perfect-square section), so the k -th largest divisor of N is N divided by the k -th smallest divisor. You only need the five smallest divisors, so factor $N = 2,014,000,000$ first.

What are the five smallest divisors of N ?

12. A divisor of $21!$ is odd exactly when it carries no factor of 2. So the probability is (odd divisors)/(all divisors); every prime except 2 cancels between the two counts, leaving only the exponent of 2 to matter. Count the factors of 2 in $21!$: the multiples of 2, then of 4, then of 8, then of 16.

How many factors of 2 does $21!$ contain?

13. Write $M + 1213 = a^2$ and $M + 3773 = b^2$. Subtracting eliminates M : $b^2 - a^2 = 3773 - 1213 = 2560$. Factor the difference of squares, $(b - a)(b + a) = 2560$. Making M as large as possible means making a as large as possible.

How should you split 2560 into $(b - a)(b + a)$ to push a as high as it goes?

Hint 2

1. The factorization is $1080 = 2^3 \cdot 3^3 \cdot 5$. The exponents are 3, 3, and 1, so the divisor count is $(3 + 1)(3 + 1)(1 + 1)$.

What does $4 \cdot 4 \cdot 2$ equal?

2. Since $588 = 2^2 \cdot 3 \cdot 7^2$, we read off $w = 2$, $x = 1$, $y = 0$, and $z = 2$. Now substitute into $2w + 3x + 5y + 7z$.

What is $4 + 3 + 0 + 14$?

3. The perfect squares up to 200 are $1^2, 2^2, \dots, 14^2$, since $14^2 = 196$ while $15^2 = 225$ is too big.

How many numbers is that?

4. The alternating sum of the digits of $\overline{1A2}$ is $2 - A + 1 = 3 - A$, which must be a multiple of 11. The only way for a single-digit A is $3 - A = 0$.

What is A , and is $\overline{132}$ really a multiple of 11?

5. The splitting $3 \cdot 3 \cdot 2$ gives exponents 2, 2, 1, which on the smallest primes is $2^2 \cdot 3^2 \cdot 5$. Compare it against the other patterns: $2^5 \cdot 3^2 = 288$ and $2^8 \cdot 3 = 768$ are both larger. What is $2^2 \cdot 3^2 \cdot 5$?

6. The smallest prime at least 50 is 53. A single prime is not allowed, and 53^2 is a perfect square (also not allowed), so the smallest legal number is the product of the two smallest such primes, $53 \cdot 59$.

What is $53 \cdot 59$?

7. A square divisor needs even exponents: 2 can be 0, 2, 4, 6, 8, 10 (six choices), 3 can be 0, 2, 4 (three choices), 5 can be 0, 2 (two choices), and 7 and 11 must both be 0. So there are $6 \cdot 3 \cdot 2 = 36$ square divisors out of $11 \cdot 6 \cdot 3 \cdot 2 \cdot 2 = 792$ total, a probability of $\frac{36}{792} = \frac{1}{22}$.

What is $m + n$?

8. There are $(10+1)(10+1) = 121$ numbers of the form $2^a 5^b$ with $0 \leq a, b \leq 10$. Exactly four of them can only be formed from two equal divisors, so they are impossible from two distinct ones: $1 = 1 \cdot 1$, $2^{10} 5^{10} = 10^5 \cdot 10^5$, $2^{10} = 2^5 \cdot 2^5$, and $5^{10} = 5^5 \cdot 5^5$.

How many remain after removing those four?

9. The 2^3 (already supplied by 2 and 4) and the prime 11 are both present once n reaches 11, but the prime 23 does not appear in the product until n reaches 23 itself, since 23 is prime and no product of smaller numbers can build it. So 23 is the bottleneck.

What is the least n ?

10. The triple $\{1, 6, 10\}$ gives $(-1)(-6)(10) = 60$ with sum $-1 - 6 + 10 = 3$, and no factor triple of 60 produces a smaller positive sum.

What is the least possible positive sum?

11. Since $2,014,000,000 = 2^7 \cdot 5^6 \cdot 19 \cdot 53$, its five smallest divisors are 1, 2, 4, 5, 8. The fifth-largest divisor is therefore $N \div 8$.

What is $2,014,000,000 \div 8$?

12. The exponent of 2 in $21!$ is $\lfloor 21/2 \rfloor + \lfloor 21/4 \rfloor + \lfloor 21/8 \rfloor + \lfloor 21/16 \rfloor = 10 + 5 + 2 + 1 = 18$. A divisor's exponent of 2 ranges over $0, 1, \dots, 18$, which is 19 equally likely values, and only the value 0 makes the divisor odd.

What is the probability?

13. Both $b - a$ and $b + a$ are even, since by the same-parity fact from Part 2 they share a parity and their product is even. So to maximize $a = \frac{(b+a)-(b-a)}{2}$ push the two factors as far apart as possible: $b - a = 2$ and $b + a = 1280$. Then $a = 639$ and $M = 639^2 - 1213 = 407108$.

What is the units digit of M ?

Greatest Common Divisor and Least Common Multiple

Introduction

The last two tools of Week 0 compare two numbers instead of examining one. Everything you learned about reading exponents transfers directly, because the greatest common divisor and the least common multiple are just the exponent-by-exponent minimum and maximum.

Example 38 (2022 AMC 10A, Problem 7)

The least common multiple of a positive integer n and 18 is 180, and the greatest common divisor of n and 45 is 15. What is the sum of the digits of n ?

- (A) 3 (B) 6 (C) 8 (D) 9 (E) 12

Solution. Factor everything in sight:

$$18 = 2 \cdot 3^2, \quad 180 = 2^2 \cdot 3^2 \cdot 5, \quad 45 = 3^2 \cdot 5, \quad 15 = 3 \cdot 5.$$

Since n divides the least common multiple 180, its only primes are 2, 3, 5, so write

$$n = 2^a 3^b 5^c.$$

The least common multiple takes the larger exponent of each prime. Comparing n with $18 = 2^1 3^2 5^0$:

$$\max(a, 1) = 2 \Rightarrow a = 2, \quad \max(b, 2) = 2 \Rightarrow b \leq 2, \quad \max(c, 0) = 1 \Rightarrow c = 1.$$

The greatest common divisor takes the smaller exponent. Comparing n with $45 = 2^0 3^2 5^1$:

$$\min(b, 2) = 1 \Rightarrow b = 1, \quad \min(c, 1) = 1 \Rightarrow c \geq 1,$$

which agrees with $c = 1$.

So $n = 2^2 \cdot 3 \cdot 5 = 60$, and the digit sum is

$$6 + 0 = \boxed{\text{(B) } 6}.$$

Remark. Every step was a statement about one prime at a time. The two conditions looked entangled in the problem statement; in the exponents they came apart into six tiny independent facts.

The main habit

Main idea/Fact (Main habit for GCD and LCM)

For small numbers, list.

For anything else, factor both numbers into primes and then work **one prime at a time**: take the **smaller** exponent for the greatest common divisor and the **larger** exponent for the least common multiple.

Common divisors and the GCD

Definition

A **common divisor** of a and b is a number that divides both. The **greatest common divisor**, written $\gcd(a, b)$, is the largest of them.

Two numbers are **relatively prime** (or **coprime**) if $\gcd(a, b) = 1$, meaning they share no prime factor at all.

Example 39

Find $\gcd(48, 180)$ by listing.

Solution. The divisors of 48 are

$$1, 2, 3, 4, 6, 8, 12, 16, 24, 48,$$

and the divisors of 180 are

$$1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90, 180.$$

The numbers on both lists are 1, 2, 3, 4, 6, 12, so

$$\gcd(48, 180) = 12.$$

Listing is fine for two-digit numbers and hopeless beyond that. Here is the method that scales.

Common multiples and the LCM

Definition

A **common multiple** of a and b is a number that both divide. The **least common multiple**, written $\text{lcm}(a, b)$, is the smallest positive one.

Example 40

Find $\text{lcm}(12, 18)$ by listing.

Solution. The multiples of 12 are 12, 24, 36, 48, ... and the multiples of 18 are 18, 36, 54, ... The first number on both lists is

$$\text{lcm}(12, 18) = 36.$$

Pitfall

Do not multiply the two numbers together and call it the least common multiple. Here $12 \cdot 18 = 216$, which is a common multiple but six times too big.

The product overshoots by exactly the greatest common divisor, as the next-but-one section explains.

By prime factorization: minimum and maximum

Main idea/Fact (The exponent rule)

Write both numbers over the same list of primes, using exponent 0 where a prime is missing:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}.$$

Then

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} \cdots p_k^{\min(a_k, b_k)}, \quad \text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdots p_k^{\max(a_k, b_k)}.$$

The reason is the divisibility rule from Part 4: a common divisor cannot exceed either exponent, so the best it can do is the minimum, and a common multiple must reach both exponents, so the least it can be is the maximum.

Example 41

Find $\gcd(48, 180)$ and $\text{lcm}(48, 180)$ by factoring.

Solution. We have

$$48 = 2^4 \cdot 3^1 \cdot 5^0, \quad 180 = 2^2 \cdot 3^2 \cdot 5^1.$$

Taking minimums prime by prime,

$$\gcd(48, 180) = 2^{\min(4,2)} 3^{\min(1,2)} 5^{\min(0,1)} = 2^2 \cdot 3 = 12,$$

which agrees with the listing above. Taking maximums,

$$\text{lcm}(48, 180) = 2^{\max(4,2)} 3^{\max(1,2)} 5^{\max(0,1)} = 2^4 \cdot 3^2 \cdot 5 = 720.$$

Pitfall

The most common error here is mixing up which operation takes which. Anchor it with sizes: the greatest common divisor *divides* both numbers, so it must be small, so it takes the minimum. The least common multiple *is divisible by* both, so it must be big, so it takes the maximum.

Exercise 42. Find $\gcd(84, 120)$ and $\text{lcm}(84, 120)$.⁵

The product identity

Main idea/Fact ($\gcd \cdot \text{lcm} = ab$)

For all positive integers a and b ,

$$\gcd(a, b) \cdot \text{lcm}(a, b) = ab.$$

The reason is one line in the exponents: for each prime,

$$\min(a_i, b_i) + \max(a_i, b_i) = a_i + b_i,$$

because the minimum and the maximum are the same two numbers in the other order.

⁵ $84 = 2^2 \cdot 3 \cdot 7$ and $120 = 2^3 \cdot 3 \cdot 5$, so the $\gcd$ is $2^2 \cdot 3 = 12$ and the lcm is $2^3 \cdot 3 \cdot 5 \cdot 7 = 840$.

Check it on the pair above: $12 \cdot 720 = 8640$ and $48 \cdot 180 = 8640$.

Pitfall

This identity is for **two** numbers only. It fails for three: with $a = b = c = 2$ the greatest common divisor is 2 and the least common multiple is 2, but the product is 8.

Exercise 43. Two numbers have product 360 and greatest common divisor 6. What is their least common multiple?⁶

Which one does the problem want?

Most of the difficulty in contest problems is not computing these; it is deciding which one is being asked for.

Main idea/Fact (Reading the story)

Reach for the **greatest common divisor** when something is being *cut up or shared out*:

- the largest identical pieces you can cut several lengths into,
- the largest number of identical gift bags several piles of candy can fill,
- the largest square tile that exactly tiles a rectangle.

Reach for the **least common multiple** when repeating events must *line up again*:

- two lights flashing at different intervals,
- runners with different lap times meeting at the start,
- the first day on which several visitors all arrive together.

The one-line test: are you *breaking something down* (gcd) or *waiting for things to sync up* (lcm)?

Example 44

A ribbon of length 48 cm and a ribbon of length 180 cm are cut into equal pieces with none left over. What is the longest possible piece, and how many pieces are there in total?

Solution. The piece length divides both ribbons, so the longest is $\gcd(48, 180) = 12$ cm. That gives $48 \div 12 = 4$ pieces from the first ribbon and $180 \div 12 = 15$ from the second, for 19 pieces in all.

Example 45

Two lighthouses flash every 48 seconds and every 180 seconds. They flash together at noon. When do they next flash together?

Solution. The next common flash is at $\text{lcm}(48, 180) = 720$ seconds, that is, 12 minutes after noon.

⁶ $360 \div 6 = 60$.

Pitfall

Same two numbers, opposite tools. Read the story, not the numbers.

A shortcut when factoring is painful

Occasionally the numbers are large and factoring them is slow. There is a trick worth knowing.

Main idea/Fact (Subtracting does not change the gcd)

For $a > b$,

$$\gcd(a, b) = \gcd(a - b, b).$$

Any common divisor of a and b also divides $a - b$, and any common divisor of $a - b$ and b divides their sum a . So the two pairs have exactly the same common divisors.

Repeating this until one number hits 0 is called the **Euclidean algorithm**.

Example 46

Find $\gcd(1001, 357)$ without factoring either number.

Solution. Subtract repeatedly, taking the whole chunk each time:

$$\gcd(1001, 357) = \gcd(287, 357) = \gcd(287, 70) = \gcd(7, 70) = 7.$$

(Here $1001 - 2 \cdot 357 = 287$, then $357 - 287 = 70$, then $287 - 4 \cdot 70 = 7$, and 7 divides 70.)

For comparison, the factorizations are $1001 = 7 \cdot 11 \cdot 13$ and $357 = 3 \cdot 7 \cdot 17$, which confirms $\gcd = 7$ but takes noticeably longer to find.

Summary

Main idea/Fact (GCD and LCM checklist) • $\gcd(a, b)$ is the largest number dividing both; $\text{lcm}(a, b)$ is the smallest number both divide.

- For small numbers, list the divisors or the multiples and read off the answer.
- Otherwise factor and work one prime at a time: minimum exponent for the gcd, maximum exponent for the lcm.
- Relatively prime means $\gcd = 1$, which means no shared prime; then $\text{lcm}(a, b) = ab$.
- $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$, for two numbers only.
- Cutting up or sharing out means gcd; repeating events lining up means lcm.
- $\gcd(a, b) = \gcd(a - b, b)$, which lets you strip large numbers down without factoring.

Problems

These are not sorted by difficulty. Try as many as you can before looking at the hints.

1. Compute $\gcd(126, 196)$ and $\text{lcm}(126, 196)$ two ways: once by listing, and once by prime factorization. Then verify your answers with the product identity.

([Hint 1](#), [Hint 2](#))

2. A florist has 84 roses, 126 tulips, and 210 daisies. She wants to make identical bouquets using all the flowers, with no flowers left over. What is the greatest number of bouquets she can make, and how many of each flower is in one bouquet?

([Hint 1](#), [Hint 2](#))

3. (2017 AMC 10A #16) There are 10 horses, named Horse 1, Horse 2, $\dots$, Horse 10. They get their names from how many minutes it takes them to run one lap around a circular race track: Horse k runs one lap in exactly k minutes. At time 0 all the horses are together at the starting point on the track. The horses start running in the same direction, and they keep running around the circular track at their constant speeds. The least time $S > 0$, in minutes, at which all 10 horses will again simultaneously be at the starting point is $S = 2520$. Let $T > 0$ be the least time, in minutes, such that at least 5 of the horses are again at the starting point. What is the sum of the digits of T ?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

([Hint 1](#), [Hint 2](#))

4. (2012 AMC 10A #16) Three runners start running simultaneously from the same point on a 500-meter circular track. They each run clockwise around the course maintaining constant speeds of 4.4, 4.8, and 5.0 meters per second. The runners stop once they are all together again somewhere on the circular course. How many seconds do the runners run?

(A) 1,000 (B) 1,250 (C) 2,500 (D) 5,000 (E) 10,000

([Hint 1](#), [Hint 2](#))

5. (2013 AMC 10A #17) Daphne is visited periodically by her three best friends: Alice, Beatrix, and Claire. Alice visits every third day, Beatrix visits every fourth day, and Claire visits every fifth day. All three friends visited Daphne yesterday. How many days of the next 365-day period will exactly two friends visit her?

(A) 48 (B) 54 (C) 60 (D) 66 (E) 72

([Hint 1](#), [Hint 2](#))

6. (2015 AMC 10B #15) The town of Hamlet has 3 people for each horse, 4 sheep for each cow, and 3 ducks for each person. Which of the following could not possibly be the total number of people, horses, sheep, cows, and ducks in Hamlet?

(A) 41 (B) 47 (C) 59 (D) 61 (E) 66

([Hint 1](#), [Hint 2](#))

7. (2016 AMC 10A #25) How many ordered triples (x, y, z) of positive integers satisfy $\text{lcm}(x, y) = 72$, $\text{lcm}(x, z) = 600$ and $\text{lcm}(y, z) = 900$?

(A) 15 (B) 16 (C) 24 (D) 27 (E) 64

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** For the listing method, write out every divisor of each number and find the largest one that appears twice. For the factoring method, get both numbers into prime powers first.

What are 126 and 196 as products of prime powers?

2. **MAIN IDEA:** Identical bouquets with nothing left over means the number of bouquets divides each of the three counts. She wants as many bouquets as possible, so this is a greatest common divisor question, not a least common multiple one.

What do 84, 126, and 210 look like in primes?

3. **MAIN IDEA:** Horse k is back at the start at time T exactly when k divides T . So you need the smallest T that is divisible by at least five of the numbers 1 through 10.

Rather than searching over T , choose five small numbers with a small least common multiple. Which five?

4. **MAIN IDEA:** Two runners are together again exactly when the gap between them has grown by a whole number of laps, so work with the difference of their speeds. There are three pairs to handle.

At what rate does the gap open between the 4.8 and 5.0 runners, and how long until that gap reaches 500 meters?

5. **MAIN IDEA:** A pair of friends coincides on the multiples of the least common multiple of their two periods. But days when all three visit are counted inside every pair, so they must be removed.

What are the three pairwise least common multiples, and what is the least common multiple of all three periods?

6. **MAIN IDEA:** Name the two independent quantities and write every other population in terms of them. Let h be the number of horses and c the number of cows.

How many people, ducks, and sheep are there in terms of h and c , and what does the grand total simplify to?

7. **MAIN IDEA:** Factor the three targets and handle the primes 2, 3, 5 completely separately, since a least common multiple condition is really one condition per prime.

The factorizations are $72 = 2^3 \cdot 3^2$, $600 = 2^3 \cdot 3 \cdot 5^2$, and $900 = 2^2 \cdot 3^2 \cdot 5^2$. What does the prime 5 force about x ?

Hint 2

1. Since $126 = 2 \cdot 3^2 \cdot 7$ and $196 = 2^2 \cdot 7^2$, take the smaller exponent of each prime for the gcd and the larger for the lcm:

$$\gcd = 2^1 \cdot 3^0 \cdot 7^1, \quad \text{lcm} = 2^2 \cdot 3^2 \cdot 7^2.$$

What are those two numbers, and does their product equal $126 \cdot 196$?

2. Factoring, $84 = 2^2 \cdot 3 \cdot 7$, $126 = 2 \cdot 3^2 \cdot 7$, and $210 = 2 \cdot 3 \cdot 5 \cdot 7$, so the common part is $2 \cdot 3 \cdot 7 = 42$. That is the number of bouquets.

Dividing each count by 42, how many of each flower goes into one bouquet?

3. Try the five smallest horses, 1, 2, 3, 4, 5, whose least common multiple is 60; that is beatable. Try 1, 2, 3, 4, 6 instead, whose least common multiple is 12, and check that no smaller T has five divisors among 1 through 10.

What is T , and what is the sum of its digits?

4. The three gap rates are 0.2, 0.4, and 0.6 meters per second, so the pairs realign every

$$\frac{500}{0.2} = 2500, \quad \frac{500}{0.4} = 1250, \quad \frac{500}{0.6} = \frac{2500}{3}$$

seconds and their whole-number multiples. You need the smallest positive time on all three lists.

Which of the answer choices is a multiple of 2500, of 1250, and of $\frac{2500}{3}$?

5. The pairs coincide every 12, 15, and 20 days, giving 30, 24, and 18 days in 365; all three coincide every 60 days, giving 6 days. Subtract the triple days from each pair count before adding.

What is $(30 - 6) + (24 - 6) + (18 - 6)$?

6. With h horses there are $3h$ people and $9h$ ducks, and with c cows there are $4c$ sheep, so the total is

$$3h + h + 9h + 4c + c = 13h + 5c.$$

Now test each answer choice: can it be written as $13h + 5c$ with h and c whole numbers?

Which choice fails for every h with $0 \leq h \leq 4$?

7. Since $\text{lcm}(x, y) = 72$ has no factor of 5, the number x has no factor of 5, so the 5^2 in $\text{lcm}(x, z) = 600$ must come from z , forcing z to contain 5^2 ; then $\text{lcm}(y, z) = 900$ is satisfied for 5 no matter what y does, so y may contain 5^0 only (since $\text{lcm}(x, y) = 72$). Handle 2 and 3 the same way, counting the allowed exponent triples for each prime, then multiply the three counts.

How many exponent choices does each of 2, 3, and 5 contribute?

Shef Scholars

Ensuring no Ramanujan or Mirzakhani is ever missed again

